EXPERIMENT 11

AIM:

To understand the creation, loading, manipulation, and basic statistical operations on DataFrames using the Pandas library.

CODE:

```
import numpy as np

import pandas as pd


# Create a DataFrame

list1 = [

   [1, "Arun", 85],

   [2, "Priya", 92]

]


df = pd.DataFrame(list1)

df.columns = ["ID", "Name", "Marks"]


print("Initial DataFrame:")

print(df)


# Display DataFrame information

df.info()


# Load student dataset from CSV file

df = pd.read_csv("student.csv")
```

```
print("\nStudent Dataset:")
print(df)

# Display DataFrame information
df.info()

# Display first five rows
print("\nFirst Five Rows:")
print(df.head())

# Display last five rows
print("\nLast Five Rows:")
print(df.tail())

# Display Marks column
print("\nMarks Column:")
print(df["Marks"])

# Display the type of Marks column
print("\nType of Marks Column:")
print(type(df["Marks"]))

# Calculate statistical measures
print("\nMean:")
```

```
print(df["Marks"].mean())

print("\nMedian:")
print(df["Marks"].median())

print("\nMode:")
print(df["Marks"].mode())

print("\nVariance:")
print(df["Marks"].var())

print("\nStandard Deviation:")
print(df["Marks"].std())

# Display descriptive statistics
print("\nDescriptive Statistics:")
print(df.describe())

print("\nDescriptive Statistics Including All Columns:")
print(df.describe(include="all"))

# Display column names
empcol = df.columns
print("\nColumn Names:")
print(empcol)
```

```
# Convert DataFrame into a NumPy array
emparray = df.values
print("\nDataFrame Values:")
print(emparray)

# Recreate DataFrame using array and column names
student_df = pd.DataFrame(
    emparray,
    columns=empcol
)

print("\nRecreated DataFrame:")
print(student_df)
```

OUTPUT:

1. Student Dataset

Output of reading student.csv:

Index	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	CSE
2	Ravi	19	76	ECE
3	Divya	20	88	EEE
4	Kiran	19	95	CSE
5	Meena	20	81	ECE
6	Ajay	19	73	EEE
7	Sneha	20	90	CSE

2. Dataset Information (df.info())

RangeIndex: 8 entries, 0 to 7

Data columns (total 4 columns):

Name — 8 non-null — object

Age — 8 non-null — int64

Marks — 8 non-null — int64

Department — 8 non-null — object

dtypes: int64(2), object(2)

memory usage: 388.0+ bytes

3. First Five and Last Five Records

df.head():

Index	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	CSE
2	Ravi	19	76	ECE
3	Divya	20	88	EEE
4	Kiran	19	95	CSE

df.tail():

Index	Name	Age	Marks	Department
3	Divya	20	88	EEE
4	Kiran	19	95	CSE
5	Meena	20	81	ECE
6	Ajay	19	73	EEE
7	Sneha	20	90	CSE

4. Selecting the Marks Column

Output of df["Marks"]:

Index	Marks
0	85
1	92
2	76
3	88
4	95
5	81
6	73
7	90

5. Descriptive Statistics for Marks

Statistic	Output
Mean	85.0
Median	86.5
Mode	73, 76, 81, 85, 88, 90, 92, 95 (all occur once)
Variance	60.57142857142857
Standard deviation	7.782764841072135

Output of df.describe():

Statistic	Age	Marks
count	8.000000	8.000000
mean	19.500000	85.000000
std	0.534522	7.782765
min	19.000000	73.000000
25%	19.000000	79.750000
50%	19.500000	86.500000
75%	20.000000	90.500000
max	20.000000	95.000000

6. Summary of All Columns (df.describe(include='all'))

Statistic	Name	Age	Marks	Department
count	8	8.000000	8.000000	8
unique	8	NaN	NaN	3
top	Arun	NaN	NaN	CSE
freq	1	NaN	NaN	4
mean	NaN	19.500000	85.000000	NaN
std	NaN	0.534522	7.782765	NaN
min	NaN	19.000000	73.000000	NaN
25%	NaN	19.000000	79.750000	NaN
50%	NaN	19.500000	86.500000	NaN
75%	NaN	20.000000	90.500000	NaN
max	NaN	20.000000	95.000000	NaN

7. Column Names

Output of df.columns:

Index(['Name', 'Age', 'Marks', 'Department'], dtype='object')

8. Dataset as a NumPy Array

Output of df.values (dtype=object):

```
[['Arun', 19, 85, 'CSE'],
 ['Priya', 20, 92, 'CSE'],
 ['Ravi', 19, 76, 'ECE'],
 ['Divya', 20, 88, 'EEE'],
 ['Kiran', 19, 95, 'CSE'],
 ['Meena', 20, 81, 'ECE'],
 ['Ajay', 19, 73, 'EEE'],
 ['Sneha', 20, 90, 'CSE']]
```

9. Reconstructed DataFrame

Output of student_df:

Index	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	CSE
2	Ravi	19	76	ECE
3	Divya	20	88	EEE
4	Kiran	19	95	CSE
5	Meena	20	81	ECE
6	Ajay	19	73	EEE
7	Sneha	20	90	CSE

RESULT:

The Pandas library was successfully used to create, manipulate, and explore a DataFrame and perform basic statistical operations.